

Reconstruction-Free Evaluation for Dead Sea Scroll Restoration: Token Alignment, Transcription-Encoded Evidence, and Unknown-Length Gaps

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Abstract

Dead Sea Scroll lacunae differ from ordinary masked-word tasks: visible letters may survive, while character length, word count, and boundaries remain uncertain. We introduce a reconstruction-free corpus derivative and two complementary evaluations that keep these information sources explicit. On 93 natural single-word lacunae, a preserved-only masked language model attains 15.1% Top-10 agreement with attributed proposals from context alone and 50.5% when supplied transcription-visible letters; a frequency baseline with the same traces reaches 36.6%. On 300 non-overlapping synthetic spans sampled across 79 held-out scrolls, the strongest unknown-length decoder reaches 7.7% exact Top-10 and recovers no three-word target. These results establish a useful but bounded conclusion: transcription evidence sharply narrows candidate sets, whereas present language models do not solve unknown-length multiword restoration. Agreement with published proposals is not historical ground truth.

1 Introduction

The Dead Sea Scrolls preserve ancient Hebrew and Aramaic in manuscripts damaged by abrasion, tears, decay, and uncertain joins. The data are not dominated by isolated missing words: in our reconstruction-free transcription derivative, only 18.8% of lacuna records occupy one source-word position, the median occupies four, and 32.7% occupy six or more. A transcription may encode partial letters and an editor’s estimate of missing extent, while the original word count and boundaries remain uncertain. A restoration system must therefore state exactly which evidence it receives.

Masked language models (MLMs) provide contextual predictions, but their tokenization can leak or obscure gap structure. A WordPiece mask normally specifies a token count, while an epigrapher observes an uncertain region measured in material space. Character models offer direct alignment, yet lexical models may rank complete words more effectively. Retrieval can expose parallels but can also leak the target or reproduce editorial traditions. Evaluation against a modern reconstruction is similarly ambiguous: it measures agreement with an edition, not recovery of inaccessible historical truth.

We organize these issues into two complementary tasks: (A) synthetic damage to preserved text with a known answer and (B) agreement with multiple attributed readings at natural lacunae. Task A hides character length, word count, and boundaries and uses exact complete-span Top-10. Task B separates context-only, visible-trace, and editor-length conditions on one fixed denominator.

The present draft makes four evidence-bounded contributions:

- a reconstruction-free non-biblical corpus derivative and a reproducible profile of 27,814 transcription-derived lacuna records;
- a shared-denominator benchmark that separates context, transcription-visible letters, and editor-derived length rather than collapsing them into a “physical” condition;
- a paired 93-target ablation with a frequency baseline, candidate-coverage reporting, manuscript-cluster uncertainty, and a composition-unseen subset; and
- a manuscript-balanced 300-span benchmark showing that unknown-length multiword decoding remains unresolved.

The contribution is therefore primarily methodological and diagnostic rather than a claim of historically correct restoration.

2 Related Work

Ancient-text restoration. Pythia introduced neural character restoration for Greek inscriptions and reported character error rate (CER) and Top-20 sequence recovery [1]. Recurrent models were also applied to fragmentary Babylonian texts [7]. Lazar et al. formulated Akkadian missing-sign completion as MLM inference and included expert evaluation [10]. Ithaca combined character and word representations with attribution, dating, saliency, and a controlled historian study [2]. Aeneas subsequently integrated images, transcriptions, parallel retrieval, arbitrary-length restoration, and historian evaluation for Latin inscriptions [3]. The field-wide survey of Sommerschild et al. emphasizes that restoration must be located within a broader scholarly workflow [17].

Semitic and manuscript models. Embible ensembles character and word transformers for synthetically masked Biblical Hebrew and Aramaic [9]. MsBERT targets orthographic variation and lacunae in post-biblical Hebrew

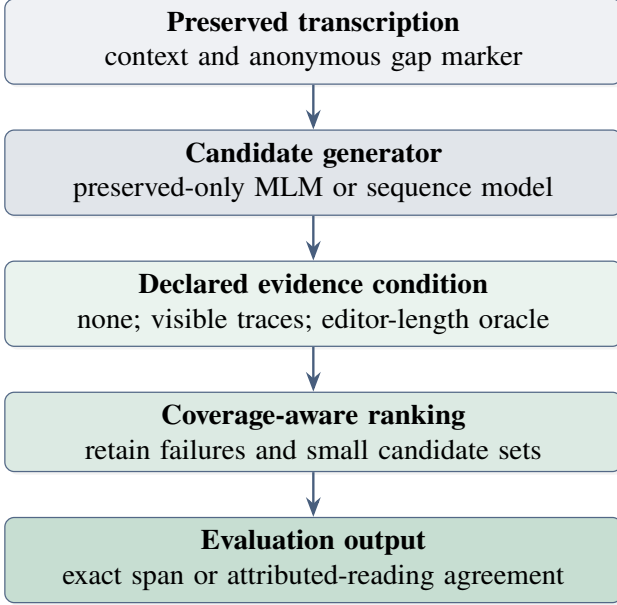


Figure 1: Evaluation architecture. Visible letters are copied from a transcription; no image or stroke inference is performed. Each result names its supplied evidence.

manuscripts [16]. TavBERT studies character-level pretraining for morphologically rich languages [13]; AlephBERT and DictaBERT provide strong Hebrew subword baselines [14, 15]. These models differ in period, register, tokenizer, and pretraining data, so a DSS comparison requires a shared target set and information condition.

Real lacunae and material evidence. The MAAT corpus preserves lacunae, restorations, and editorial structure from EpiDoc instead of reducing every example to random masking [8]. Levine et al. evaluate ranked, variable-length completions for Coptic manuscripts and explicitly frame predictions as scholar assistance [11]. For the DSS, Uzan et al. restore damaged letter images with a modified Pixel-CNN [19]. That image-level task is complementary to our transcription-level span ranking; claims about ink or strokes require an explicit multimodal connection between them.

3 Tasks and Information Conditions

Let $x_{<g}$ and $x_{>g}$ be preserved context around a gap g . A system returns an ordered set $C_k(g)$ of at most k complete strings. For synthetic damage with known preserved target y_g ,

$$H_k(g) = \mathbb{I}[y_g \in C_k(g)]. \quad (1)$$

For a natural lacuna with a set Y_g of distinct, bibliographically attributed proposals,

$$A_k(g) = \mathbb{I}[C_k(g) \cap Y_g \neq \emptyset]. \quad (2)$$

A_k is *literature agreement*; neither inclusion nor exclusion establishes truth.

We distinguish four information conditions:

- **CONTEXT:** linguistic context only.

Table 1: Reconstruction-free non-biblical corpus profile. Held-out refers to the checkpoint-associated split used by the preserved-only models.

Measure	Full corpus	Held-out
Scroll identifiers	736	220
Training/evaluation chunks	1,647	407
Preserved word tokens	95,736	25,313
Lacuna records	27,814	6,814
Damaged word positions	165,239	43,401

- **VISIBLE:** candidates must contain the Hebrew segments shown outside reconstruction brackets, respecting anchored edges when encoded.
- **EDITORLENGTH:** candidate length must fall within one character of the QD display/initial length. This is an oracle/editor-assisted diagnostic.
- **VISIBLE+LENGTH:** the conjunction of the preceding filters, also oracle-assisted because of length.

Visible traces are transcription evidence, not independent observations collected by our system. The current pipeline neither reads multispectral images nor estimates stroke probabilities.

4 Data

Corpus construction. The corpus is derived from the ETCBC DSS Text-Fabric dataset [6]. Biblical scrolls are excluded. Editorial reconstruction text is never emitted as a training label: physically preserved characters remain visible and gaps become anonymous <GAP> tokens. Deterministic scroll-level partitions contain 1,002 training, 238 development, and 407 held-out chunks. Corpus and lacuna files are recorded with SHA-256 hashes. Table 1 reports counts regenerated directly from the validated derivative.

Scroll eligibility is task-specific. An archival registry and a modeling population answer different questions. We retain all 736 non-biblical scroll identifiers for provenance, including fragments that contribute only damage meta-data. A scroll enters primary MLM text only if it yields at least one 128-unit chunk with 20 preserved words. Consequently, 377 scrolls contribute primary text and 359 do not (Table 2). The latter are not deleted: 25 contain no strictly preserved words, while 71 satisfy the predeclared extreme-fragmentation flag of fewer than 20 preserved source words and a preserved-position fraction below 10%. These overlapping groups remain auditable but are excluded from primary linguistic modeling and contextual-restoration target selection. Neither evaluation contains an extreme-fragmentation scroll. Preservation fraction here is a property of classified transcription positions, not a measurement of surviving parchment area.

This policy avoids a global intactness blacklist. A globally damaged manuscript may contain a locally usable island; such material may enter a separately reported preserved-run ablation if it satisfies reconstruction-free local criteria, but

Table 2: Task-specific scroll eligibility. Categories overlap: zero-preserved and extreme-fragmentation scrolls are subsets of the no-chunk population. Preservation fraction is a transcription-position proxy, not parchment-area intactness.

Population	Scrolls	Treatment
Archival registry	736	retain for provenance
Primary MLM contributors	377	at least one eligible chunk
No primary MLM chunk	359	exclude from MLM text
Zero preserved words	25	exclude from linguistic modeling
Extreme fragmentation	71	flag; exclude from primary context task

Table 3: Lacunae by estimated source-word positions. Counts derive from consecutive damaged positions after editorial text is removed; they are structural estimates, not direct measurements of a single physical hole.

Positions	Full corpus	Held-out
1	5,239 (18.8%)	1,207 (17.7%)
2	4,536 (16.3%)	1,090 (16.0%)
3	3,997 (14.4%)	962 (14.1%)
4–5	4,941 (17.8%)	1,186 (17.4%)
6+	9,101 (32.7%)	2,369 (34.8%)

it cannot silently enter the primary population. Conversely, a high scroll-wide preservation rate cannot rescue a target with inadequate local left/right context. Target eligibility is therefore determined at the lacuna as well as the scroll level.

What does a lacuna look like? The 27,814 records contain 165,239 damaged source-word positions (Table 3). Their median span is four positions, the 90th percentile is 13, and the maximum is 268. These are consecutive source-word positions after editorially supplied text has been removed. They describe the structure of the transcription and must not be read as 27,814 independent, metrically measured physical holes.

The transcription nevertheless preserves useful evidence. At least one word retains a visible trace in 23,632 lacuna records (85.0%), but only 49,130 of the 165,239 damaged word positions (29.7%) retain any trace. An estimated missing-character extent is available for 25,509 records (91.7%); its median is five characters and its nearest-rank interquartile range is 2–12. Both left and right preserved context are available for 26,467 records (95.2%), with a median of 14 total context words. The contrast between record-level evidence and position-level sparsity motivates constrained ranking without assuming that every hidden word has a trace.

Two evaluation views. No single benchmark answers both whether a system can recover a known string and whether it can surface a defensible proposal at a genuine lacuna. We therefore use the complementary views in Table 4: synthetic damage supplies an observable answer under controlled masking, while the QD set preserves natural ambiguity and supports only literature-agreement claims.

Table 4: The two principal evaluation views. Synthetic damage has an observable preserved answer; QD measures agreement with attributed proposals and not historical truth.

Set	Targets	Scrolls	Reference
Unknown-length synthetic	300	79	preserved complete span
QD natural lacunae	93	40	184 attributed readings

Synthetic alignment diagnostic. The shared 729-word diagnostic masks intact content words and tests whether candidate generation aligns with the same character interval. It is useful for tokenizer coverage, but it does not simulate unknown-length natural damage and is not the primary benchmark.

Unknown-length spans. We hide 300 preserved DSS spans, balanced across one, two, and three words, after fitting decoding parameters on 60 development spans. Character length, word count, and boundaries are hidden. To prevent large compositions from dominating, deterministic round-robin sampling cycles through manuscripts and rejects overlapping hidden intervals. The test covers 79 held-out scrolls (77, 55, and 41 in the respective word-count strata) and uses two context words on each side.

Qumran Digital targets. The QD snapshot dated 21 May 2026 contains 1,811 publication rows at 346 candidate locations. Predeclared structural and context filters retain 93 non-biblical single-word targets across 40 scrolls and 184 distinct attributed readings. Crucially, a proposal is not removed merely because it conflicts with a tested evidence filter; it remains a possible reference and becomes unretrievable in that condition. All 93 targets are held out under the frozen manifest used for checkpoint training. QD selected disputed sites and is working scholarly data; the benchmark is neither random nor exhaustive.

5 Models and Decoding

Word-span baseline. The word-span baseline enumerates one to three word-mask hypotheses and ranks complete surfaced strings. It is fine-tuned only on preserved non-biblical text. The tokenizer can represent 81.7% of complete synthetic targets with one token per word; all other targets remain in the denominator as misses.

Character and byte models. TavBERT supplies character-aligned masks and ByT5 supplies a byte-level sequence-to-sequence comparison [12]. Character alignment avoids WordPiece drops, but alignment alone does not imply superior exact-span ranking. Three ByT5 checkpoints vary only the training seed (41–43) and share three epochs, batch size 32, learning rate 10^{-3} , and the same corpus hash.

Embibble-style combinations. We implement both the paper-described overlap rule and a development-fitted rank fusion of word and character candidates. Under the locked tie-break, a simpler system is retained when exact Top-10

Table 5: Shared synthetic alignment diagnostic ($n = 729$ preserved content words). Hit rates use all targets as the denominator; alignment failures are misses.

System	Hit@1	Hit@10	Drops
TavBERT base (character)	7.27%	21.12%	0.00%
TavBERT FT (character)	7.96%	20.58%	0.00%
MsBERT FT (WordPiece)	6.04%	13.31%	38.27%
ByT5 unified (byte)	0.82%	5.35%	0.00%

ties. This prevents selecting a more complex ensemble on secondary metrics alone.

Frequency baseline. For the QD task, we rank Hebrew surface forms by frequency in preserved training chunks, then apply exactly the same trace and length filters as the MLM. This baseline isolates how much apparent success follows from a narrow constraint set rather than contextual modeling.

6 Evaluation Protocol

The primary automatic task is exact complete-span Top-10 under unknown length, word count, and word boundaries. Exact Top-1/5/20, MRR, CER, boundary F1, word-count mean absolute error, decoder failure, calibration, selective accuracy, and latency are secondary. Slot-level word hits are diagnostic only.

The frozen split is scroll-disjoint. Test data are never used for selection. Failed decoding, tokenizer misalignment, unrepresentable targets, empty candidate sets, and lists shorter than K remain in the denominator. Exact duplicate normalized chunks are absent across the development/test boundary. Maximum preserved-word five-gram Jaccard is 0.174; none of 407 held-out chunks reaches 0.5. Because 26 of 88 composition labels cross partitions, we separately report the QD targets whose composition labels are unseen in train and development.

We report 95% bootstrap intervals clustered by scroll and paired cluster intervals for evidence-condition deltas. Target-level predictions, exclusions, manifests, and code/data/model/result hashes accompany the aggregate tables. ByT5 uses three matched seeds; the MLM and TavBERT comparisons remain single-checkpoint analyses and are labeled accordingly. These choices follow reproducibility guidance in NLP [5, 18].

7 Results

The results follow the task’s sources of uncertainty rather than a leaderboard: we first test whether tokenization can align a known interval, then require exact recovery when length and boundaries are hidden, and finally measure agreement with attributed proposals at natural lacunae. Metrics are comparable only within each table.

Table 6: Unknown-length synthetic damage on 300 non-overlapping preserved DSS spans sampled across 79 held-out scrolls. Intervals use 10,000 manuscript-cluster resamples. The oracle uses unavailable gold word length.

System	Top-1	Top-5	Top-10	95% CI	CER
Preserved-only word span	4.0%	6.7%	7.7%	5.0–10.8%	0.893
Preserved-only TavBERT	1.0%	2.3%	2.7%	0.9–5.1%	0.898
Emble-style overlap	2.3%	2.3%	2.3%	0.6–4.6%	0.884
Dev-fitted rank fusion	4.3%	7.0%	7.7%	5.1–10.6%	0.868
Oracle word-length filter	6.7%	10.7%	12.0%	–	0.860

Table 7: Exact Top-10 and representability by hidden word count. Each stratum has 100 spans. Coverage is diagnostic; uncovered cases remain misses.

Measure	1 word	2 words	3 words
Word-span exact Top-10	22.0%	1.0%	0.0%
Complete-span coverage	91.0%	77.0%	77.0%

Alignment is necessary but insufficient. Table 5 uses a genuinely shared denominator. TavBERT base obtains the highest Hit@10, while DSS fine-tuning does not improve it. MsBERT’s aligned-only rates are 9.78% Hit@1 and 21.56% Hit@10, but presenting those alone would condition on successful tokenization. With 279 of 729 failures counted as misses, its all-target rates are 6.04% and 13.31%. DictaBERT-char is omitted because the available 22.80% result comes from a different 250-word pilot.

Unknown-length multiword restoration remains open.

The word-span model and rank fusion tie at 7.7% exact Top-10 (Table 6); the simplicity rule retains the word model. The oracle word-length filter reaches 12.0%, quantifying the benefit of unavailable structural information rather than a deployable gain. Table 7 exposes the decisive weakness: the baseline falls from 22.0% on one-word spans to 1.0% on two-word spans and 0.0% on three-word spans. Thus the benchmark rejects, rather than validates, a claim of solved unknown-length restoration.

Three matched ByT5 checkpoints were rerun on the identical target manifest. Their overall exact Top-10 scores are 1.3%, 1.3%, and 1.0%; every checkpoint obtains 0% on the two- and three-word strata. The seed result therefore confirms a stable negative finding rather than a competitive sequence baseline.

Visible transcription traces carry most of the QD signal.

On the same 93 targets, visible-trace filtering changes MLM Top-10 agreement from 15.1% to 50.5% (Table 9). The paired improvement is 35.5 points, with a 24.5–46.5 manuscript-cluster 95% interval. The absolute visible-trace score has a 40.3–60.0% interval. Under the same traces, the frequency baseline reaches 36.6%; the paired MLM advantage is 14.0 points with a 3.7–23.1 interval. Constraint narrowing therefore explains part, but not all, of the result. Coverage is imperfect: the MLM filter returns no candidate for 13 targets and fewer than ten for 37, all retained as misses. The editor-derived length proxy adds no Top-10 gain over visible traces alone and is not used in the headline.

Table 8: Unknown-length ByT5 replications on the same 300 targets. All three checkpoints use identical training hyperparameters; seed is the only varied training setting.

Seed	Epochs	Batch	LR	Top-1	Top-10
41	3	32	1e-03	0.0%	1.3%
42	3	32	1e-03	0.7%	1.3%
43	3	32	1e-03	0.0%	1.0%

Table 9: Agreement with any attributed reading at 93 Qumran Digital targets. The visible-trace MLM Top-10 scroll-cluster bootstrap 95% CI is 40.3–60.0. All rows retain the same targets and references. Editor-derived length is reported only as an oracle-assisted condition.

Condition	Top-1	Top-5	Top-10	Top-20
MLM: context only	4.3%	11.8%	15.1%	17.2%
Frequency: context only	0.0%	2.2%	3.2%	6.5%
MLM: visible traces	33.3%	48.4%	50.5%	54.8%
Frequency: visible traces	22.6%	32.3%	36.6%	36.6%
MLM: editor length (oracle)	6.5%	14.0%	16.1%	18.3%
MLM: traces + length (oracle)	32.3%	49.5%	50.5%	53.8%

Trace-count error analysis. Visible conditioning is not uniformly beneficial. For the 12 targets with only one transcribed letter, Top-10 falls from 41.7% context-only to 16.7%; for 14 two-letter targets it rises from 28.6% to 57.1%, and for 67 targets with at least three letters it rises from 7.5% to 55.2%. Single-letter placement is therefore too brittle to treat as a reliable hard filter in this sample. A production system should soften uncertain traces and expose conflicts rather than silently discarding candidates.

Composition-unseen analysis. The frozen split is manuscript-disjoint but not composition-disjoint. Restricting QD to 24 targets from 11 scrolls whose composition labels do not occur in train or development yields 16.7% context-only and 62.5% visible-trace Top-10; the matched frequency baseline reaches 33.3%. This small, selected subset is a robustness check, not a substitute for training on a composition-grouped split.

8 Comparative Synthesis

Table 10 places the framework in its lineage. Lazar establishes ancient Semitic MLM completion; Embible motivates granularity ensembles; MsBERT motivates manuscript-domain pretraining; Aeneas demonstrates arbitrary-length multimodal restoration. Our contribution is a DSS corpus and evaluation framework that separates supplied evidence from model ability; we do not claim a new state-of-the-art architecture.

9 Reproducibility and Data Statement

Core paper-facing numbers are regenerated from a frozen result snapshot containing target-manifest, model, input,

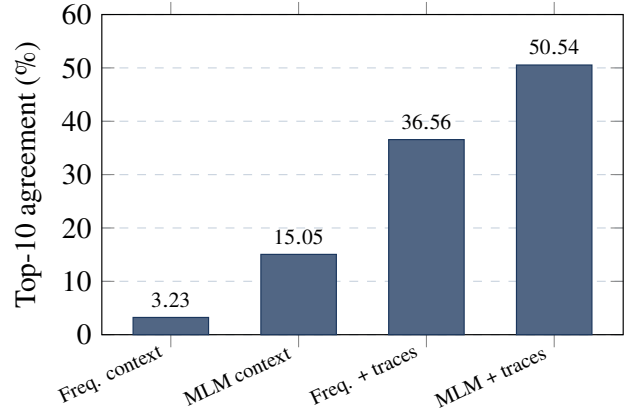


Figure 2: QD Top-10 literature agreement on a fixed 93-target denominator. Trace filters help both systems; the MLM retains a 14.0-point advantage over frequency under the same visible evidence.

and artifact hashes; a machine-readable table specification records evidence class and sources. The snapshot reproduces the 300-span baselines and three matched ByT5 evaluations on one target hash, and reruns QD over 93 target-level records. The preserved corpus validator enforces non-biblical membership, removal of modern reconstruction text, anonymous gap representation, and disjoint scrolls. A separate audit records composition overlap, exact duplicates, and five-gram similarity.

The corpus manifest records primary-population counts and per-scroll preserved/damaged statistics for every extreme-fragmentation flag. Target-level predictions include candidate counts, evidence-condition ranks, and attributed readings, allowing every aggregate to be recomputed without rerunning a model.

The artifact records the ETCBC Text-Fabric version, QD snapshot date, source licenses, checkpoint hashes, training epochs, batch sizes, learning rates, beam sizes, seeds, software environment, and sample hashes. Experiments ran locally on an Apple M4 Max with 36 GB unified memory using MPS. Public redistribution remains subject to source-corpus and checkpoint licenses.

10 Limitations and Responsible Use

Synthetic damage supplies observable answers but differs from natural deterioration and editorial uncertainty. QD targets cover selected disputed sites, not a representative random sample, and its attributed proposals are incomplete. Text-Fabric editorial labels can encode modern reconstructions and therefore cannot serve as physical truth. Transcription-derived traces inherit editorial judgments. Period, dialect, genre, scribal practice, formulaic language, and orthography may all confound apparent model quality.

Eligibility thresholds also induce selection effects. Requiring 20 preserved words per chunk favors scrolls with sufficiently dense local text, while a contiguous-run rule favors uninterrupted passages and suppresses the damaged contexts encountered at inference. The extreme-fragmentation

Table 10: Methodological synthesis. Scores are not compared across corpora. A dash indicates that a dimension is not central to the cited evaluation.

Dimension	Lazar (2021)	Embible (2024)	MsBERT (2024)	Aeneas (2025)	This work
Material/language	Akkadian tablets	Biblical Heb./Aram.	Hebrew MSS	Latin inscriptions	DSS transcriptions
Generation unit	subword/sign	word + char.	WordPiece	character sequence	word span + char. ablation
Natural lacuna eval.	expert sample	no	yes	yes	QD literature agreement
Unknown length	greedy masks	masked spans	missing word	arbitrary length	primary protocol
Material evidence	transcription	mask extent	transcription	image + text	traces; editor-length oracle
Primary caution	domain/task scope	synthetic damage	token alignment	edition-derived targets	no physical truth

flag is an operational audit threshold rather than a claim that a manuscript is philologically worthless. Results therefore generalize to eligible transcription contexts, not automatically to the most severely damaged physical fragments.

The QD model is a single training seed; its interval captures target and manuscript sampling, not optimization variance. The composition-unseen subset is small, and composition labels are incomplete. The synthetic benchmark covers only one to three words and two context words per side. Character models solve alignment but have not solved exact multiword generation. Direct claims about ink require image inputs and image/transcription registration, which are absent here.

The system is intended for decision support. Every candidate should retain its model score, evidence condition, rejected constraints, provenance, and uncertainty. Suggestions must not be silently inserted into editions, used to overwrite attributed readings, or presented as recovered historical text. Scholars remain responsible for paleography, material joins, syntax, genre, parallels, and publication decisions.

11 Conclusion

Transcription-visible letters substantially narrow candidate space, and a contextual MLM improves over frequency under the same evidence. This is the paper’s positive result. Its equally important negative result is that token alignment and granularity ensembling do not solve unknown-length multiword restoration: exact recovery approaches zero beyond one word. Shared denominators, coverage accounting, manuscript-balanced sampling, clustered uncertainty, and explicit oracle labels make both conclusions reproducible without turning literature agreement into historical truth.

Responsible NLP Research Checklist

This draft follows the ARR checklist structure [4]. Data provenance, language, licensing, exclusions, reconstruction redaction, compute, statistical units, and limitations are stated above. This work contains no human-participant study.

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